

Chapter 2: National and Regional Trends and Forecasts

Introduction

This section examines the region's population, employment, and activity density; evaluates socioeconomic growth trends and forecasts within TCAMPO, the Commonwealth of Virginia (Commonwealth), and the United States of America; and maps demographic patterns within the Tri-Cities. Understanding where the region's existing residents and jobs are concentrated and where they are growing is critical for identifying the region's current and emerging transportation needs. Additionally, Environmental Justice (EJ) analysis of the region's demographic patterns provides insight into equity considerations, helping to identify areas that may face transportation disadvantages due to race, income, limited English proficiency, household vehicle availability, disability, or age, helping to ensure equitable access to multimodal transportation options throughout the region.

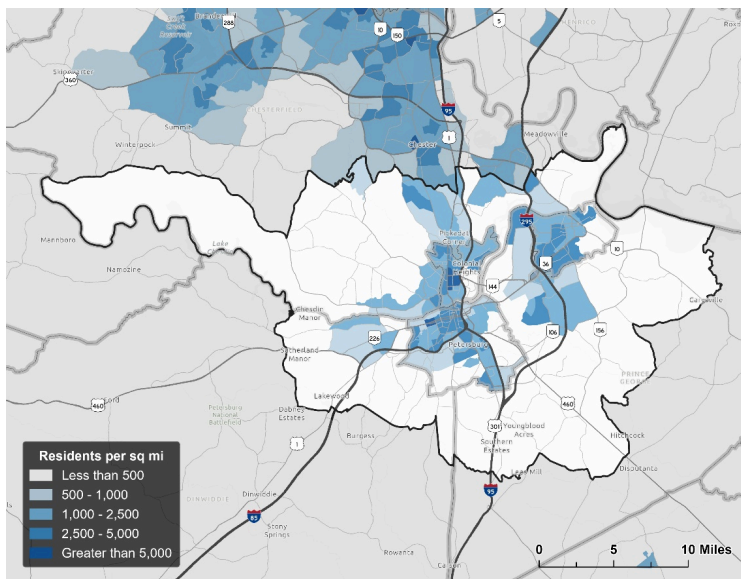
Because Chesterfield, Dinwiddie, and Prince George extend beyond the TCAMPO boundary, block group, tract, and county-level data from various sources were adjusted using area-weighted interpolation to reflect only the portion of these jurisdictions within TCAMPO. Geographic units located entirely within the boundary, such as census blocks and Traffic Analysis Zones (TAZ), were included without adjustment.

Population, Employment, and Activity Density

The following maps illustrate existing population, employment, and activity density by block group in the Tri-Cities region and beyond. Population estimates are drawn from the American Community Survey (ACS) 2023 5-Year Estimates. Employment estimates are drawn from the 2023 Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES) Workplace Area Characteristics (WAC) dataset.

Figure 3 shows population density (residents per square mile) across the region. Population density is highest in Petersburg, Colonial Heights, and Hopewell, with additional concentrations in Chesterfield, Dinwiddie, and Prince George Counties, particularly along the I-95 and I-85 corridors and near major institutions such as Virginia State University (VSU) and Fort Lee. Notable exceptions to this pattern exist within eastern Hopewell and southwestern Petersburg, where lower population densities are due to industrial, commercial, and park land uses rather than rural or undeveloped land.

Figure 3. TCAMPO Population Density by Block Group

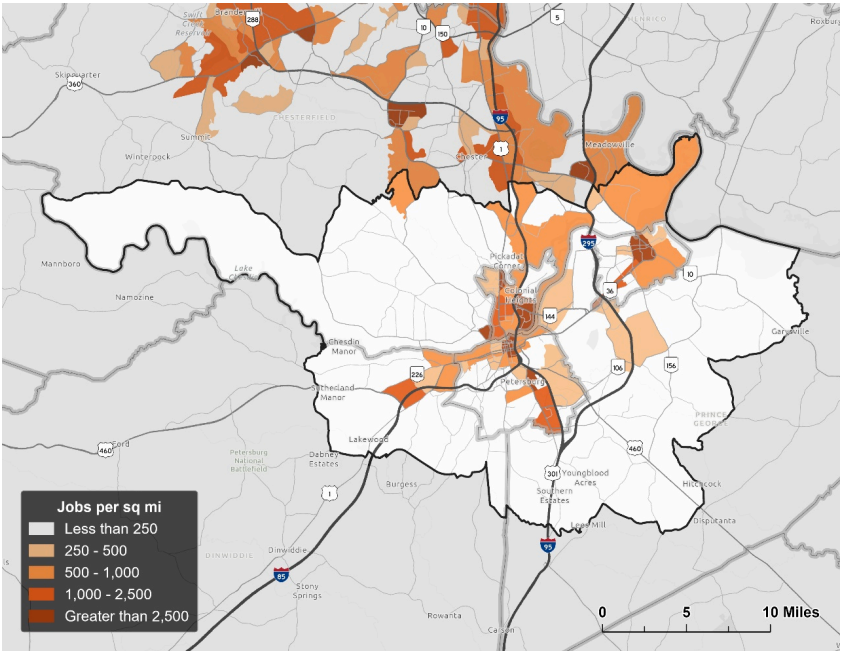


Source: 2023 ACS 5-Year

Figure 4 illustrates the region's employment density (jobs per square mile). Employment is primarily concentrated along the I-95 and I-85 corridors through Dinwiddie, Colonial Heights, and Petersburg, with additional dense clusters just north of the TCAMPO boundary in Chester and Meadowville. Other areas of high employment density are evident in the historic downtown cores of Petersburg and Hopewell, near VSU in Ettrick, at Southpark Mall in Colonial Heights, and along arterial principal arterial corridors such as South Crater Road (US-301), Oaklawn Boulevard (VA-36), and East Hundred Road (SR-10). Rural portions of the region, particularly to the west, south, and east, show minimal employment density. Overall, employment distribution in the region is tied to interstate access, and major commercial and

institutional centers. Due to limitations in federal jobs reporting within the LEHD dataset, employment associated with Fort Lee is not captured in this map.

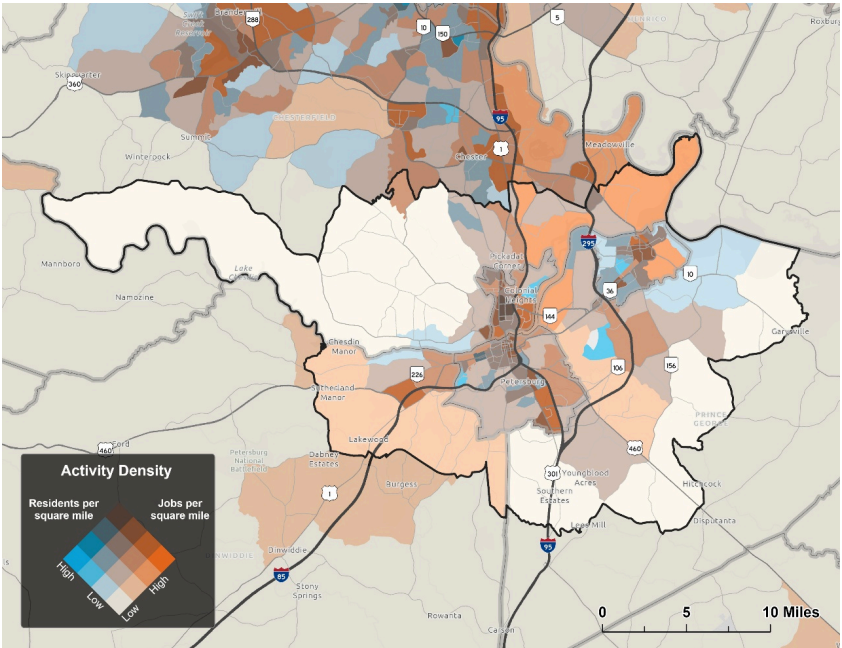
Figure 4. TCAMPO Employment Density by Block Group



Source: 2023 LEHD

Figure 5 combines the two measures above to show activity density (residents plus jobs per square mile) across the region. Activity density is highest where population and employment concentrations overlap, most notably in Colonial Heights, Hopewell, and Petersburg and along the I-95 and I-85 corridors. High activity is also illustrated near VSU and Fort Lee. As with population and employment densities individually, activity density is lowest in the rural portions of the region to the west, south, and east.

Figure 5. TCAMPO Activity Density by Block Group



Source: 2023 ACS 5-Year, 2023 LEHD

Population, Household, and Employment Trends

Historical population and household trends from 1990 to 2020 were analyzed using decennial census block-level data obtained from the Integrated Public Use Microdata Series (IPUMS) National Historical Geographic Information System (NHGS). Employment trends for the Tri-Cities region were estimated using county-level annual averages from the Bureau of Labor Statistics (BLS) Quarterly Census of Employment and Wages (QCEW) during the same period. Population and employment growth supports economic development. It also directly affects travel demand, leading to increased congestion, emissions, and greater crash frequency due to higher traffic volumes. These factors indicate that as a region grows, there is a clear need for higher investment in a robust multimodal transportation system, including

pedestrian and bicycle safety improvements, increased transit accessibility, and reductions in highway congestion, as well greater destination access for all modes of transportation.

As shown in **Figure 6**, the population of the Tri-Cities region grew steadily between 1990 and 2020, increasing by 24.4 percent from 134,954 to 167,920. Over the same period, population growth in both Virginia and the United States outpaced that of the region, with Virginia's population increasing by 39.5 percent and the nation's population increasing by 33.3 percent.

Figure 6. Regional, State, and National Population Growth by Decade, 1990-2020

Source: IPUMS National Historical Geographic Information System

Figure 7 illustrates that population growth has been uneven across the six jurisdictions and the three decennial periods between 1990 and 2020. During this period, the region's largest population growth has occurred in counties dominated by suburban and rural areas. The TCAMPO portion of Chesterfield County grew by 69.3 percent between 1990 and 2020, experiencing the largest decennial increases between 1990-2000 and 2000-2010. The TCAMPO portion of Dinwiddie County also experienced strong growth, increasing by 58.3 percent over the same period, although growth slowed considerably after 2010. The TCAMPO portion of Prince George County grew by approximately 56 percent between 1990 and 2020, with its largest increase occurring between 2010 and 2020. Colonial Heights experienced steady population growth in each decade between 1990 and 2020 resulting in a 13.1 percent overall increase. In contrast, Petersburg experienced a 12.8 percent population decline between 1990 and 2020, with a 3 percent increase since 2010. During this period, Hopewell's population remained stable with only a 0.3 percent decline between 1990 and 2020, although its population has slightly recovered in the last two decennial periods.

Figure 7. TCAMPO Population Growth by Jurisdiction and Decade, 1990-2020

Source: IPUMS National Historical Geographic Information System

Figure 8 shows that the number of households in the region grew by 28.4 percent between 1990 and 2020, increasing from 48,586 to 62,362. Like population growth during this same period, households grew at a slower rate than both Virginia (44.9 percent) and the nation as a whole (37.9 percent). Furthermore, this growth was not evenly distributed among the region's jurisdictions studied in this analysis. Chesterfield County experienced the highest growth (74.2 percent), followed by Dinwiddie County (61.1 percent) and Prince George County (46.9 percent). In contrast, household growth was positive, but substantially less in Colonial Heights (20.0 percent), Hopewell (4.2 percent), and Petersburg (4.4 percent). The latter two are notable given that Petersburg's and Hopewell's population actually declined over this same period.

The relationship between population and household growth rates is reflected in changes in household size. Across all jurisdictions, average household size declined by 3.1 percent from 2.8 to 2.7 people per household between 1990 and 2020. This reduction is consistent with broader demographic trends toward smaller households across the Commonwealth (minus 3.7 percent) and the United States (minus 3.4 percent). Within the region, household size decreased in all jurisdictions except Prince George, where it increased from 3.4 to 3.6 people per household. In contrast, Petersburg experienced the largest decline, falling by 16.5 percent from 2.6 to 2.2 people per household.

Figure 8. Regional, State, and National Household Growth by Decade, 1990-2020

Source: IPUMS National Historical Geographic Information System

As with population growth, **Figure 9** illustrates that household growth varied significantly by jurisdiction and decennial period. In Chesterfield, Dinwiddie, Prince George, and Colonial Heights, household growth was strongest between 1990 and 2000. Hopewell and Petersburg followed the opposite pattern, with both cities recording their fastest household growth between 2010 and 2020. After losing households throughout the 1990s and 2000s, Petersburg had the highest household growth rate of any TCAMPO jurisdiction between 2010 and 2020, even as its population growth remained the second lowest in the region. This discrepancy suggests that much of Petersburg's new housing stock was absorbed by smaller households.

Figure 9. TCAMPO Household Growth by Jurisdiction and Decade, 1990-2020

Because the earliest available year for LEHD data is 2002, historical block-level employment data is not available for this analysis. Therefore, QCEW county-level annual averages were proportionally allocated to the Tri-Cities boundaries using area weighted interpolation based on block-level total employment in the 2023 LODS WAC dataset. **Figure 10** shows that employment in the Tri-Cities region has increased by about 47.3 percent from 37,198 in 1990 to 54,780 in 2020. In contrast with population and households, employment is growing at a faster rate than the state (32.3%) and the country (28.1%). Employment growth was comparatively slower between 2000 and 2010, reflecting the impacts of the Great Recession and broader economic conditions that reduced job growth during that period.

Figure 10. Regional, State, and National Employment Growth by Decade, 1990-2020

Source: Bureau of Labor Statistics QCEW County-Level Annual Averages

Shift-Share Employment Growth by BLS Supersector

In this section, employment growth in the Tri-Cities region is evaluated across all BLS supersectors using shift-share analysis. Supersectors are high-level industry groupings used by BLS for the QCEW program, aggregating employment across related two-digit NAICS sectors to provide a wider view of economic activity in an area. These industries can be further grouped into the Goods-Producing and Service-Providing domains. **Figure 11** illustrates the NAICS sectors that comprise each super-sector and domain.

As shown in **Figure 12**, net employment in the Tri-Cities region increased across most major industry supersectors between 2019 and 2023, although the magnitude and causes of growth varied by sector. Shift-share analysis decomposes a regional employment change into three components: the National Growth Component (NGC), which measures growth attributable to the overall national employment trends; the Industrial Mix Component (IMC), which reflects the industry-specific national growth patterns; and the Local Share Component (LSC), which captures growth resulting from factors unique to the region. Combined, the NGC, the IMC, and the LSC help identify the industries driving regional employment growth and determine whether observed changes reflect broader national trends or a competitive advantage specific to the Tri-Cities region.

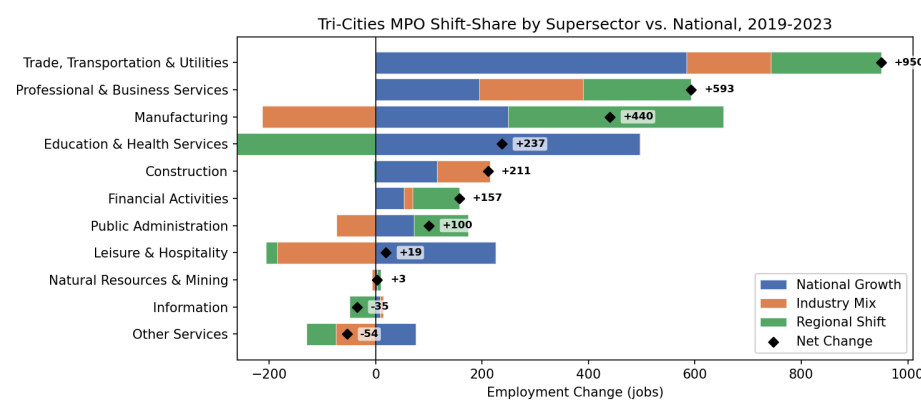
Figure 11. QCEW High-Level Industry Crosswalk

Domain | Super-Sector NAICS-Sector

The analysis revealed that Trade, Transportation & Utilities was the region's largest source of net employment growth between 2019 and 2023, adding 950 jobs. Most of that growth came from the NGC (+585 jobs), or what the region would have gained/lost in this industry if it simply grew at the national economy's overall rate. The IMC (+158 jobs) and LSC (+208 jobs) were both smaller than the NGC but positive, suggesting this industry grew faster than the overall national economy and the region also outperformed the national trend for this specific industry.

The strong positive local growth in both Trade, Transportation & Utilities and Manufacturing is consistent with the region's position along the I-95 and I-85 corridors and its growing role as a goods movement and freight logistics hub, reflected in warehousing, distribution, and freight-dependent manufacturing activity in the Tri-Cities region. The Professional & Business Services, Public Administration, and Financial Activities supersectors also had strong local growth and outperformed national growth for these industries. Because freight-dependent sectors in the Goods-Producing and Service-Providing domains are outperforming national trends at the local level, freight infrastructure including highway capacity, interchange access, and rail/intermodal connections along the I-95 corridor will be increasingly important to sustaining this growth going forward.

Figure 12. TCAMPO Shift-Share by Supersector vs. National, 2019-2023

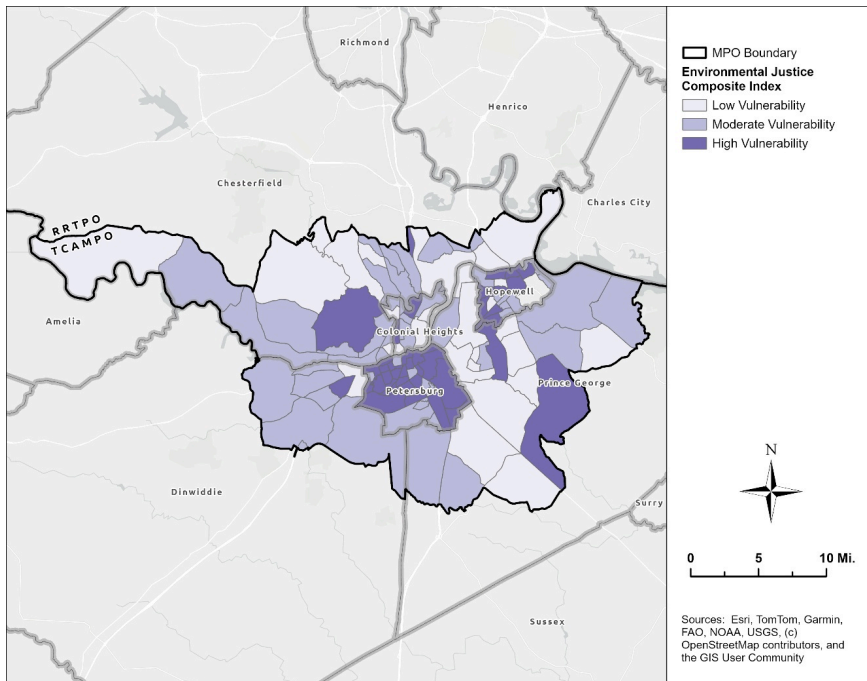


Demographic Profiles

This section summarizes the Environmental Justice (EJ) analysis which is incorporated into TCAMPO's prioritization process for projects considered for inclusion in Plan2050. The analysis designates EJ Areas of Concern, defined as census block groups having relatively higher concentrations of residents or households that may have limited access to viable transportation choices. Identifying areas where these populations are concentrated helps ensure that transportation investments are planned and prioritized in a way that promotes equitable access across the region.

For this analysis, six EJ indicators are derived from 2023 ACS 5-Year Estimates. Each indicator addresses a distinct dimension of vulnerability relevant to transportation access and investment equity: minority and low-income populations, persons with limited English proficiency, persons with disabilities, elderly persons, and households without access to a vehicle. These population groups were selected based on federal Title VI considerations, as they may experience have fewer transportation options, exhibit greater dependence on transit, or require additional accommodations to access employment, healthcare, education, and other essential services. Because raw percentages for each of the six indicators vary in scale and distribution, each indicator is reclassified into 1-3 (lowest to highest EJ Concern) ordinal vulnerability score using tertile breaks derived from the Tri-Cities block group distribution. The six reclassified indicator scores are summed to produce a composite vulnerability index, which maps the Composite EJ Index. Because each reclassified factor ranges from 1 to 3, the composite index has a potential range of 6 (all indicators at lowest vulnerability) to 18 (all indicators at highest vulnerability). The equal weighting of all six factors reflects the absence of a defensible empirical basis for differential weighting. As shown in **Figure 13**, the composite vulnerability index is reclassified into a final three-level Environmental Justice Composite Index, representing relative concentrations of EJ-related vulnerability across the Tri-Cities region.

Figure 13. Composite EJ Index by Block Group



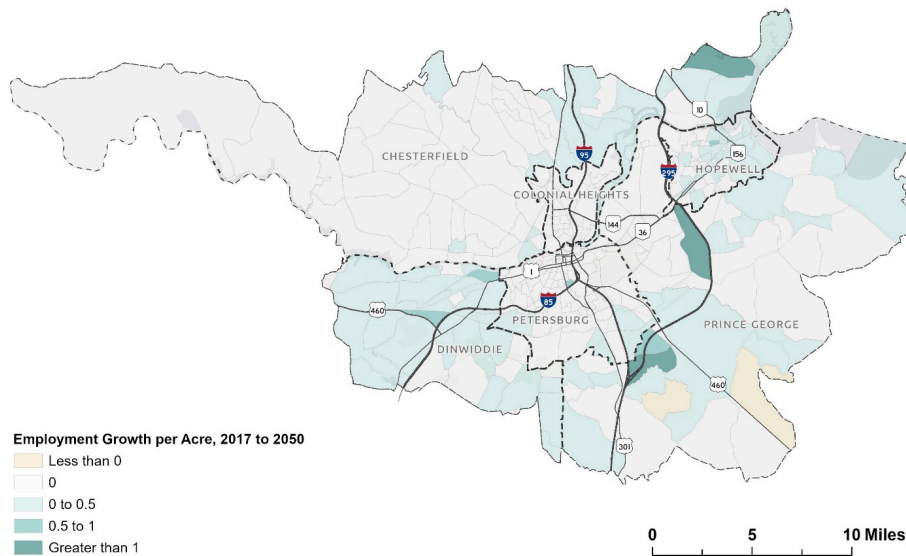
Socioeconomic Forecasts

The MPO developed 2017-2050 socioeconomic forecasts (population, housing, autos, school enrollment, and employment) by TAZ to support the Richmond/Tri-Cities (RTC) Travel Demand Model, which TCAMPO and RRTPO use for long-range planning and corridor studies. Note that forecasts were assigned to TAZs across all of Dinwiddie County and Prince George County, not just the portions within the TCAMPO planning area. Across the six jurisdictions, total population is projected to grow by about 20.0% from 191,300 to 229,600; households (occupied housing units) by about 30.9% from 64,400 to 84,300, and total employment by about 22.9% from about 62,400 to 76,800 by 2050. As shown in **Table 1**, growth is uneven across jurisdictions. Chesterfield and Prince George are expected to see the largest growth (Prince George's households will nearly double, up 88.5%), while Dinwiddie, Hopewell, and Petersburg are expected to grow more slowly, and Colonial Heights' employment is projected to stay flat.

Table 1. 2017-2045 Socioeconomic Forecasts

Jurisdictions	Total Population		Households		Total Employment	
	2017	2050	2017	2050	2017	2050
Chesterfield	43,683	61,806	14,909	21,403	10,384	16,645
Dinwiddie	28,504	30,477	9,980	10,500	8,647	10,438
Prince George	45,881	60,888	12,543	23,640	13,790	19,422
Colonial Heights	17,395	19,646	6,782	7,682	9,344	9,344
Hopewell	22,941	23,471	8,910	9,325	7,137	7,366
Petersburg	32,918	33,280	11,280	11,727	13,130	13,540
TCAMPO	191,322	229,568	64,404	84,277	62,432	76,755

Figure 14. Employment Change per Acre by Traffic Analysis Zone (TAZ), 2017-2050

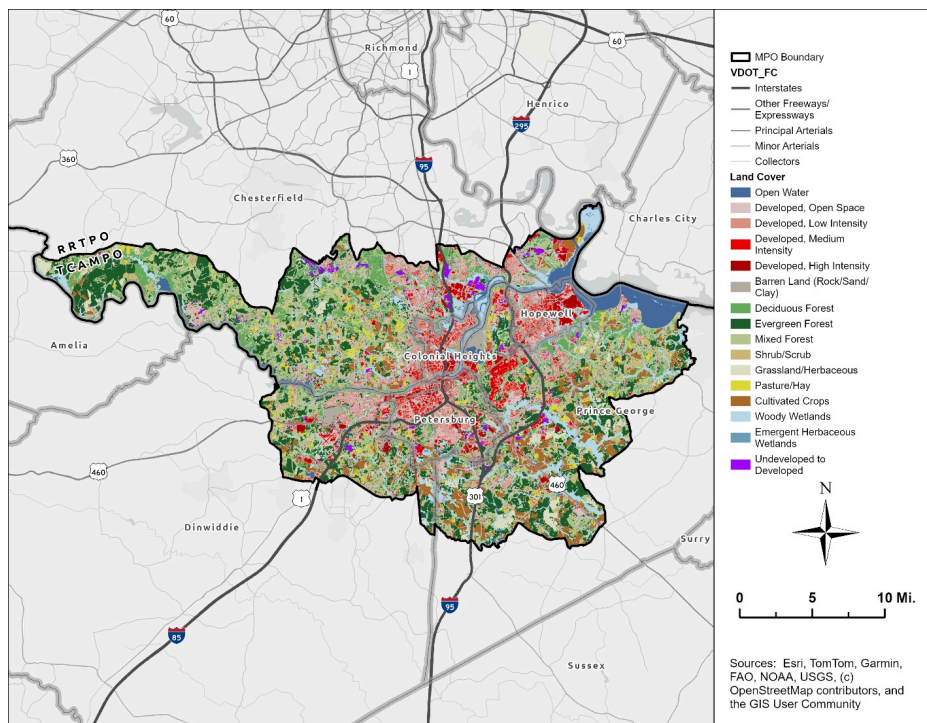


Land Use and Development

Land use in the Tri-Cities region is characterized by a wide range of land cover classes, varying from high intensity developed areas to undeveloped land cover classes such as forests, cultivated areas, and wetlands. As shown in **Figure 15**, which shows land cover change between 2014 and 2024 using data from the National Land Cover Database (NLCD), the region's most developed areas are located in the urbanized cores of Petersburg, Colonial Heights, and Hopewell, as well as the region's major transportation corridors. The NLCD is a national land cover dataset produced from satellite imagery and provides detailed classifications of both developed and undeveloped land across the United States.

The map below highlights areas that remained developed or undeveloped in 2024, as well as locations where undeveloped land transitioned to developed land. These development changes are primarily concentrated along existing growth corridors and at the urban fringe. Areas that experienced little or no change are generally associated with established urban centers, agricultural lands, forested tracts, and environmentally sensitive areas such as wetlands and riparian corridors.

Figure 15. Existing Land Cover and Transitions from Undeveloped to Developed Areas, 2014-2024



Source: U.S. Geological Survey (USGS), 2014 and 2024, Annual NLCD Collection

Figure 16 illustrates the current extent of developed land across the region's six jurisdictions. The chart below summarizes, for the portions of each jurisdiction within the Tri-Cities MPO, the percentage of land that is developed, undeveloped, or changed from undeveloped to developed between 2014 and 2024. To highlight regional growth patterns, individual land cover classes were consolidated into broader “developed” and “undeveloped” categories for this analysis.

This chart shows that region’s urban core – Colonial Heights, Hopewell, and Petersburg – has more than 60 percent of total land developed, while Dinwiddie, Prince George, and Chesterfield each has less than 30 percent developed. Across the Tri-Cities MPO as a whole, about 28 percent of land is developed, suggesting a region defined by a densely developed core surrounded by lower-density, largely undeveloped areas. While the conversion of undeveloped to developed land is not directly tied to long-range transportation planning needs, this trend mirrors the region’s broader patterns of population, household, and employment growth. Petersburg’s comparatively high rate of land conversion reflects both its population growth during the last decade and its relatively small size and limited remaining undeveloped land. The areas in Petersburg that transitioned from undeveloped to developed land are concentrated along Wells Road in south Petersburg, and along the US-301/I-95 corridor near the Bon Secours Medical Center and the Wagner Road interchange (Exit 48).

Figure 16. Percentage of Land that is Developed, Undeveloped, and Transitioned to Developed Use, 2014-2024

Source: U.S. Geological Survey (USGS), 2014 and 2024, Annual NLCD Collection

Travel Patterns

Population and employment changes shape both development and travel patterns throughout the region. Understanding these existing patterns provides a basis for developing future transportation needs. Examining trip flows, household vehicle availability, mode share, and forecasted travel demand helps inform decision makers about the investments that are needed to support economic vitality, improve mobility, and enhance quality of life for residents and visitors alike. Figure X illustrates how residents currently travel throughout the region, showing the distribution of trips across different modes of transportation.

Trips by Mode - TBD

Household vehicle availability is one of the biggest determinants of travel behavior and mode choice, shaping whether a trip is made by car, transit, walking, or not at all. **Table 2** presents the distribution of households by vehicle availability and jurisdiction. Vehicle availability is categorized by the number of vehicles available per household, ranging from zero vehicles to three or more vehicles. Because census tracts are the smallest geographic unit for which vehicle availability data are available from the American Community Survey (ACS), estimates for the portions of Chesterfield, Dinwiddie, and Prince George within TCAMPO were calculated using population-weighted spatial interpolation.

As shown below, vehicle availability varies considerably across the region, generally tracking differences in population density and household income. Petersburg has the highest share of zero-vehicle households (14.84%) and the highest combined zero-and-one-vehicle share (63.27%). Hopewell also shows an above-average zero-vehicle share (11.66%). At the other end, Dinwiddie has the highest share of three-plus-vehicle households (46.34%), reflecting its more rural, auto-dependent character, with Prince George and Chesterfield showing similar trends. Overall, TCAMPO (8.20% zero-vehicle) skews slightly higher than Virginia statewide (6.10% zero-vehicle) in low-vehicle-availability households, driven mainly by Petersburg and Hopewell.

Table 2. Household Vehicle Availability by Jurisdiction

Jurisdiction	Households	No Vehicle	1 Vehicle	2 Vehicles	3+ Vehicles
Chesterfield	16,017	4.22%	22.22%	40.93%	32.63%
Dinwiddie	4,409	5.90%	19.37%	28.42%	46.34%
Prince George	10,204	3.47%	22.95%	34.08%	39.50%
Colonial Heights	7,530	6.79%	33.73%	35.96%	23.52%
Hopewell	9,255	11.66%	35.73%	26.12%	26.49%
Petersburg	15,138	14.84%	48.43%	25.84%	10.89%
TCAMPO	62,553	8.20%	31.87%	32.49%	27.45%
Virginia	3,365,732	6.10%	31.13%	37.25%	25.52%

Source: 2024 ACS 5-Year

In addition to household vehicle availability, mode share is determined by other factors including trip purpose and trip distance or travel time. **Figure 17** shows that the majority of workers who are 16 years and over and who did not work from home drove alone to work across all commute durations, accounting for 82% of all commuters and more than 85% of commuters in travel times between 15 minutes to one

hour. Walking was most prevalent among the shortest commutes, comprising nearly 24 percent of workers with travel times under 10 minutes, and declining sharply for commutes of 20 minutes or more. Carpooling represented a small but consistent share across all time intervals, ranging from approximately 7 to 11 percent. Transit use was minimal overall but was most notable among the longest commuters, representing about 10 percent of workers with travel times of 60 minutes or more.

Figure 17. Mode Share by Travel Time, TCAMPO Region

Source: 2024 ACS 5-Year Estimates

Changes in Traffic Congestion (2020-2025) - TBD

County to County Commuter Flows

The Tri-Cities region is part of an interconnected metropolitan area that extends beyond its borders. Examining commuting patterns both within the region and beyond provides insight into the interconnectedness of the region's economy and the role of the transportation network in supporting those connections.

Table 3 summarizes county-to-county commuter flow data derived from 2025 StreetLight Origin-Destination traffic volumes (Average Daily Traffic) and Home-Based Work (HBW) trip purpose data. This 2025 traffic volume dataset covers all days of the week using default time periods, and includes the TCAMPO-area portions of Chesterfield, Dinwiddie, and Prince George Counties, along with the Cities of Colonial Heights, Hopewell, and Petersburg. External volumes were derived using Traffic Analysis Zones (TAZs), including zones outside the Tri-Cities Region from the RTC model, as well as other origin and destination trips outside the RTC model.

As shown in Figure 18, the share of Internal-Internal (I-I) trips that begin and end within TCAMPO varies considerably by jurisdiction, ranging from a high of 71.7 percent in Prince George to a low of 37.6 percent in Chesterfield. Prince George, Colonial Heights, Petersburg, and Hopewell have the greatest share of internal trips, with all but Hopewell retaining more than two-thirds of commute trips within the MPO. Prince George also has the highest proportion of trips that begin and end within the same jurisdiction (45.4 percent), reflecting the influence of Fort Gregg-Adams as a major local employer and trip generator.

In contrast, nearly two-thirds of commute trips associated with the TCAMPO portion of Chesterfield travel outside the MPO. Much of this travel is to destinations within the Richmond region, particularly the portions of Chesterfield County located within the Richmond Regional TPO, as well as Henrico and the City of Richmond. This pattern reflects the strong economic and commuting relationship between the Tri-Cities area and the greater Richmond metropolitan region.

Figure 18. Share of TCAMPO Trips Staying Within the Region, by Jurisdiction

Source: StreetLight, 2025

Among internal TCAMPO flows, the strongest connections are centered on Prince George and Petersburg (Table 3). Prince George exchanges the highest volume of trips with any other jurisdiction in the region, particularly with Petersburg (2,203 trips) and Hopewell (2,086 trips), reflecting its high levels of population and employment and its role as the geographic center between the region's urban cores. Chesterfield also shows a notable bidirectional relationship with Prince George, Colonial Heights, and Petersburg (approximately 600–700 trips in both directions), which is consistent with Chesterfield's overall size growing employment in both jurisdictions and Petersburg's role as an activity center for surrounding jurisdictions.

Table 3. Commute Flows within Tri-Cities Area MPO

		TCAMPO Destinations					
		Chesterfield	Colonial Heights	Dinwiddie	Hopewell	Petersburg	Prince George
TC A M P O O r i g i n s	Chesterfield	1,562	654	312	319	618	733
	Colonial Heights	606	1,850	263	271	506	684
	Dinwiddie	283	265	427	91	535	326
	Hopewell	312	321	101	2,036	319	1,052
	Petersburg	628	507	586	301	2,155	1,106
	Prince George	625	671	250	1,034	1,097	6,340

